Model Paper - Database Management Systems

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Q1

Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university is managing its academic departments and student enrollments in a database system. The university has several entities, including the 'Department', which has attributes such as 'DepartmentID', 'DepartmentName', and a composite attribute 'Location' that includes 'BuildingName' and 'RoomNumber'. Each department offers multiple 'Courses', characterized by attributes 'CourseID', 'CourseName', and 'Credits'. Students, captured in the 'Student' entity with attributes 'StudentID', 'FullName', and a multivalued attribute 'EmailAddresses', enroll in courses through an 'Enrollment' relationship, which records 'EnrollmentDate'. Additionally, the university categorizes students into subtypes based on their academic level; 'UndergraduateStudent' includes unique attributes like 'Major' and 'YearOfStudy', while 'GraduateStudent' has its own attributes, such as 'ThesisTitle' and 'AdvisorName'. The cardinality between 'Department' and 'Course' is (1,N), while 'Course' to 'Enrollment' is (1,N), and 'Student' to 'Enrollment' is (1,N)

Figure: EER

a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly.

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer.

Q2

Consider a relation R(A, B, C, D, E) with the following set of functional dependencies F over R: F={{AB->C, C->D, DE->A, A->B, B->E}}. In this schema, A and B are composite keys representing entities The given functional dependencies show relationships between attributes that suggest complex dependencies, making normalization essential to achieve a proper database schema.

a) Find all the keys in relation R using the attribute closure method.

b) Is R in 3NF? Give reasons for your conclusion.

c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations.

Q3

A university is developing a comprehensive database system to manage its academic records. The system includes various entities such as Students, Courses, Instructors, and Enrollments. Each Student has attributes including student ID, name, age, and major. Courses have attributes like course ID, title, credit hours, and semester offered. Instructors possess attributes such as instructor ID, name, and department. An Enrollment records the relationship between Students and Courses with additional attributes for the grade received. The university wants to establish constraints to ensure data integrity and enforce business rules like 'A student can enroll in a maximum of five courses per semester.' The database should also cater to queries concerning student performance across semesters and aggregate data effectively.

a) Write SQL queries to in type 2 driver, jdbc api calls are converted to native java api calls. accept or refute the above statement justifying your answer.

b) String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used.

c) Briefly explain the options available for transferring data between two sources such as between tables and servers.

d) Briefly explain how authentication and authorization are achieved in SQL Server.

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments

database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication.

ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role.

iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role.

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases.

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases.

Q4

Consider the following schema of a database designed for a hospital management system: Patient (patientId: int, name: varchar(100), dob: date, address: varchar(150)), Doctor (doctorId: int, name: varchar(100), specialty: varchar(50), phone: varchar(15)), Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)). The 'Patient' table stores information about patients including personal details such as name, date of birth, and address. The 'Doctor' table contains records of doctors including their names, specialties, and contact number. The 'Appointment' table records the appointments made by patients with doctors, including the appointment status and date. Each patient can have multiple appointments with different doctors. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. Fee (feeId: int, patientId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following:

i. Find the name and phone number of the doctor who has the specialty 'Cardiology'.

ii. Find the patient with the highest aggregate value using GROUP BY on related appointment records.

iii. Find the name and name from Doctor, Patient, and Appointment where Appointment.appointmentDate is not null.

b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status.

c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'.

Q5

Consider the following relational database schema containing airline flight information.

Here the passenger relation gives the details of the passengers who book flights.

The agency relation keeps the details of agents who book flights for passengers.

The flight relation stores the details of each available flight.

The booking relation stores required booking details.

passenger (pid, pname, pgender, pcity)

agency (aid, aname, acity)

flight (fid, fdate, time, departs, arrives)

booking (pid, aid, fid, fdate)

a) Display the titles of the books printed in 2019.

b) Display the titles of the books published in New York city.

c) Write a relational algebra expression using aggregation/grouping for the required computation.

d) Write a relational algebra expression using set difference or division for the stated condition.

e) Display the titles of books with most number of authors.

Q6

Consider a relation R(BookID, Title, AuthorID, AuthorName, PublisherID) with the following set of functional dependencies F = { BookID -> Title, AuthorID -> AuthorName, PublisherID -> PublisherName, AuthorID, PublisherID -> Title }. This schema represents a library database system managing book information, authors, and publishers, requiring normalization.

a) Identify the candidate key(s) for relation R and justify your answer.

b) Determine the highest normal form that relation R satisfies and explain your reasoning.

c) Perform the normalization process on relation R up to third normal form (3NF) and provide the final decomposed relations.